

Categorization of Urdu News Articles Using Machine Learning Models

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December 8, 2024

Abstract

The increasing availability of news articles on the internet has created a need for automatic categorization to facilitate information retrieval and aggregation. Categorizing Urdu news articles into relevant topics can help readers access the most relevant news and enable better recommendations. This report presents the results of a machine learning project aimed at categorizing Urdu news articles into predefined categories. Three machine learning models were employed: Bayesian, Neural Network (NN), and Logistic Regression. The models were trained and evaluated on a dataset of 1140 news articles scraped from three popular Urdu news websites: Express, Geo, and Jang. After preprocessing, data cleaning, and analysis, the models were evaluated using various metrics such as accuracy, and classification reports. Among the models tested, Logistic Regression outperformed the others, achieving the highest accuracy. This study highlights the challenges of text classification in a language with unique syntactic and morphological features, and the importance of selecting the right model for handling such complexities.

1 Introduction

The categorization of news articles is an essential task in Natural Language Processing (NLP) and has several real-world applications, including news aggregation, content recommendation, and information retrieval. In this project, we focus on the task of categorizing Urdu news articles using machine learning models. The objective is to compare the performance of three models: Bayesian, Neural Network (NN), and Logistic Regression, in classifying Urdu news into multiple categories. This report details the data collection process, the methodologies employed for model training, the findings, and the challenges encountered during the project.

2 Methodology

2.1 Data Collection

A total of 1140 articles were scraped from three major Urdu news websites:

- <https://www.express.pk>
- <https://urdu.geo.tv>
- <https://jang.com.pk/category/latest-news>

The data was collected using web scraping techniques, with around 80 articles per category from Express News, 90 articles per category from Jang, and 60 articles per category from Geo. The articles were collected across multiple categories such as Politics, Sports, Entertainment, Science and Technology, etc.

2.2 Data Preprocessing and Cleaning

The scraped data was exported into a CSV file, which included columns for titles, URLs, content, and gold labels (categories). The preprocessing steps were as follows:

- Removal of punctuation, numbers, and unidentifiable characters.
- Removal of entries with NULL values.
- Standardization of Urdu alphabet for consistency.
- Distribution analysis of the dataset for identifying imbalances.

2.3 Data Analysis and Distribution

The analysis of the dataset revealed the following insights:

- **Distribution of Articles Across Categories:** The dataset did not contain any imbalance in the number of articles per category.
- **Length Distribution:** The article lengths showed a skewed distribution, with some articles being exceptionally short or long. This prompted the removal of outlier articles based on length, particularly those outside the 5th and 95th percentiles.
- **Content Distribution:** The contents of the articles were analyzed for frequency distribution, which helped in understanding the types of vocabulary used across different categories. This analysis aided in fine-tuning the preprocessing and vectorization steps.

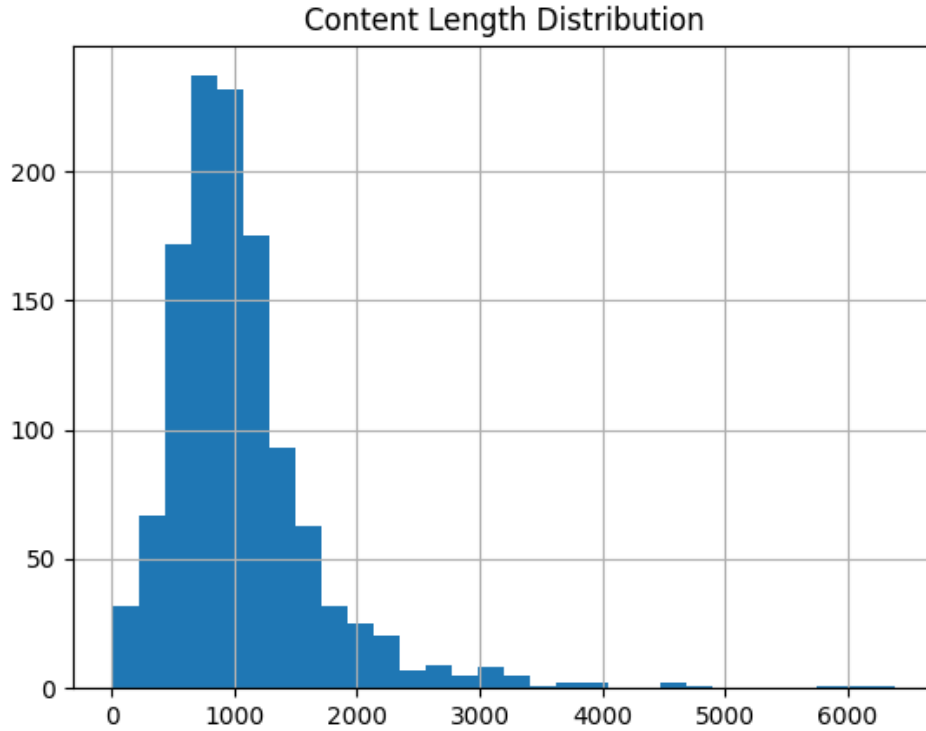


Figure 1: Content Distribution across Different Categories

2.4 Data Splitting and Vectorization

Across all models, the data was randomly split into a training set (80%) and a testing set (20%) to ensure diversity.

2.5 Model Description

Three machine learning models were trained for classification:

- **Bayesian Model:**
 - A Naive Bayes model was implemented, i.e., it was assumed that the probability of each word is independent of all the other words.
 - A matrix A was created to store the frequency of each word in each class.
 - Priors for each class were calculated by dividing the frequency of each class by the total training points.

- Likelihoods were calculated using the word frequency matrix A . Add one Laplace smoothing was used to avoid zeros.
- For each test point, the log probability of that point being in each class was calculated. This was done using:

$$\log P(c | x) = \log P(c) + X \cdot \log(P(x | c))$$

where X is a vector of weights assigned to each word by the TfidfVectorizer, x is the input vector and c is the class. The use of weights slightly improved our results and also took care of new words that were absent in the training set.

- Once the probability of the input vector to be in each class was calculated, hard max was used to make the final prediction.
- The model was evaluated using `accuracy_score` and `classification_report` from sklearn.

• Neural Network Model:

- Punctuations and numbers were removed, and UrduHack stopwords were used to clean the text.
- The content and titles were vectorized separately with different optimized parameters such as n-gram size and maximum length of vector. These were then merged using numpy's `hstack` into a single feature for training the models.
- Outlier articles were removed based on length (articles shorter than the 2.5th percentile and longer than the 97.5th percentile).
- The dataset was resampled using SMOTE to handle class imbalance.
- A three-layer Neural Network with 'Adam' optimizer and moderate dropout values was used.
- Relatively optimal parameters like epochs and batch size were selected along with dynamic learning rate to smooth the accuracy curve.

• Logistic Regression Model:

- A vocabulary corpus was created using a custom BagofWords class using manually implemented tokeniser.
- The frequency of each word in vocabulary corpus was treated as a feature.
- The categorical labels for sports, world, entertainment, science-technology, and business were replaced with numerical labels.

- Batch gradient descent was used in the training process, combined with Elastic Net regularization, with $\lambda = 0.0001$ and a learning rate of 0.1.
- Epochs were hypertuned to 180 in order to minimize training and validation losses. Although higher values of epochs produced a comparable amount of loss, they were deliberately avoided to avoid overfitting.
- A one-vs-all approach was adopted by creating five classifiers, each dedicated to one of the relevant categories. For each classifier, the corresponding category was assigned a label of 1, while all other categories were changed to 0.
- In the testing phase, all the new words that did not appear in the training phase were discarded.
- The `evaluate` function returned the probability of a label being equal to 1 (i.e., the probability of the data point belonging to that category). This process was repeated for all categories, and the resulting logits were stored in a 2D numpy matrix. The maximum logit for each instance was selected using a hardmax approach. Finally, a classification report and a confusion matrix were used to evaluate the performance of the classifier.

3 Findings

The performance of the three models was evaluated using accuracy scores and classification reports. Although all the models consistently provided an accuracy higher than 90%, there were notable distinctions in the model, the way they see the data and make assumptions about the underlying distributions that they are trying to model. The results are as follows:

- **Logistic Regression:** Achieved the average accuracy of 93%.
- **Neural Network:** Achieved an average accuracy of 92% and highest of 94%.
- **Bayesian:** Achieved an average accuracy of 91% Although accurate, the model performed less effectively compared to the other models due to the simplistic nature of the approach despite some amount of feature engineering. Another reason that might have caused the model to perform poorly can be the huge assumption that the probability of each word is independent of the others. Also to be noted is the small size of the training set, which might have given us inaccurate priors that do not represent real word class frequencies.

3.1 Archetype Data Points in a Category

A notable issue observed in the categorization process across all models was the misclassification of articles from the 'Science and Technology' category. Many of these articles were mistakenly classified into other categories due to their overlap in content. Articles in 'Science and Technology' often contained discussions on tech innovations, health, or research that were similar to those found in other categories like 'Health' or 'Business'. This overlap made it difficult for the models to accurately distinguish between these classes. As a result, these articles were frequently misclassified into categories that shared similar vocabulary or context. The consistent misclassification of 'Science and Technology' articles affected the overall accuracy of the models, with a noticeable decrease in performance. This issue was particularly prominent in the Neural Network model lowering its accuracy.

	precision	recall	f1-score	support
business	0.92	0.94	0.93	47
entertainment	0.98	0.98	0.98	46
science-technology	0.85	0.89	0.87	44
sports	0.98	1.00	0.99	47
world	0.92	0.85	0.89	55
accuracy			0.93	239
macro avg	0.93	0.93	0.93	239
weighted avg	0.93	0.93	0.93	239

Figure 2: Classification metric for Logistic Regression

This figure shows the misclassification of "Science and Technology" articles by Logistic Regression. While the overall stats are strong, the performance for this category is notably poor, highlighting an issue with category overlap. This was the case with all models.

3.2 Semantics of Models

The Bayesian model assumes the independent nature of tokens in the distribution and assigns probability of a token being a part of a certain class without considering its correlation and dependence with tokens that are near it. Because of

this reason, it fails to comprehensively capture the underlying distribution of articles across categories. The Logistic Regression model and Neural Network model outperformed Bayesian due to their ability to handle feature selection and regularization more effectively, particularly in a multi-class setting. While both Neural Network and Logistic Regression models were supposed to be the most accurate models, with the small training dataset size of 1000, the Logistic Regression model worked the best.

4 Limitations and Conclusion

4.1 Limitations

There were several limitations encountered during the project:

- Small training data. Only around 1000 articles were scraped and 800 of them were used for training that made models, especially Neural network, suffer the most as it relies on a large learning dataset to accurately tune the weights.
- Variability in article length and content quality made it challenging to maintain consistency across categories.
- The models performed inconsistently for a specific category, and more advanced supervised tuning and bigger training data could have improved results.

4.2 Conclusion

In conclusion, while all three models performed well with accuracy above 90%, Logistic Regression proved to be the most effective due to its simplicity and the ability to handle class imbalances and feature selection effectively, especially with small training data. Future improvements could include the use of more advanced deep learning models or further data augmentation to address the class imbalances, overlaps and mistaken misrepresentations.